Canonical Glossary

Reconciled Computational Interaction Science vocabulary

Version 1.0.0-integration.1

Governance

This document owns reconciled canonical wording for the integration release. It does not introduce new theory. Source papers remain provenance and evidence.

Domain commitments

A set of commitments about admissible entities, categories, relations, identity conditions, modalities, and constraints in a domain. In computational work, these commitments may be approximated through ontologies, schemas, contracts, types, or model definitions, but none of those artifacts is automatically a philosophical ontology.

Boundary: Domain commitments specify what the model admits. They do not by themselves assign meaning, produce a representation, or determine interpretation.

Canonical ID	domain-commitments
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	ontology commitments
Deprecated	None
Dependencies	None
Originating paper	Ontology, Semantics, Representation, and Interpretation
Supporting papers	Ontology, Semantics, Representation, and Interpretation; CAN INTERACTION BECOME A PRIMARY SCIENTIFIC OBJECT?
Contracts	None

System state

The physical, computational, social, or modeled condition of a declared system at a particular time.

Boundary: State is not synonymous with data, meaning, representation, or observation. A state becomes content-bearing only under a specified semantics.

Canonical ID	system-state
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	state
Deprecated	None
Dependencies	Domain commitments, State transition
Originating paper	Opportunity, Affordance, and Action Availability
Supporting papers	Opportunity, Affordance, and Action Availability; Ontology, Semantics, Representation, and Interpretation; Interaction; Interaction Strategies; Predicting Interaction
Contracts	Projection Contract, Opportunity Contract, Interaction Contract, Behavioral Evidence Contract, Measurement Contract, Similarity Contract, Transfer Contract, Strategy Contract, Prediction Contract

Semantic assignment

A rule-governed assignment of content, denotation, truth conditions, inferential consequences, or transition behavior to expressions, states, signs, or operations within a specified system and domain.

Boundary: Formal semantic assignment is distinct from a person’s interpretation or understanding.

Canonical ID	semantic-assignment
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently

Measurement	Measure only under declared operationalization
Aliases	formal semantics
Deprecated	None
Dependencies	Domain commitments
Originating paper	Ontology, Semantics, Representation, and Interpretation
Supporting papers	Ontology, Semantics, Representation, and Interpretation
Contracts	None

Semantic state

The content-bearing assignments and relations currently licensed by a specified semantics for a system or discourse state.

Boundary: No observer-independent semantic state is assumed without an explicit theory and scope.

Canonical ID	semantic-state
Maturity	Proposed
Observable	No - latent/inferred
Latent	Yes
Measurement	Requires latent-variable model
Aliases	None
Deprecated	None
Dependencies	Semantic assignment
Originating paper	Ontology, Semantics, Representation, and Interpretation
Supporting papers	Ontology, Semantics, Representation, and Interpretation; Representation, Invariance, and Projection; Computational Interaction Science
Contracts	None

Information

A typed measure or characterization of distinctions, correlations, descriptions, or content relative to a specified source, code, semantic relation, probability model, or use.

Boundary: Information is not a synonym for stored variables, data, meaning, semantic state, or interpretation. Every claim that information is preserved must name the information theory, source, code, task, or relevance criterion.

Canonical ID	information
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	None
Dependencies	None
Originating paper	Ontology, Semantics, Representation, and Interpretation
Supporting papers	Ontology, Semantics, Representation, and Interpretation; Representation, Invariance, and Projection; Interaction Strategies; Predicting Interaction; INTERACTION AS A SCIENTIFIC EXPLANANDUM
Contracts	None

Representation

A materially or formally realized vehicle that provides selective access to a declared target through a mapping or correspondence and an interpretive practice.

A representation may preserve, transform, omit, or introduce structure.

Boundary: A representation is not identical with its target. Equal reference does not imply semantic, inferential, behavioral, or interactional equivalence.

Canonical ID	representation
Maturity	Proposed

Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	representational vehicle
Deprecated	None
Dependencies	Projection
Originating paper	Representation, Invariance, and Projection
Supporting papers	Representation, Invariance, and Projection; INTERACTION AS A SCIENTIFIC EXPLANANDUM; Ontology, Semantics, Representation, and Interpretation; Interaction Transfer; Computational Interaction Science
Contracts	None

Example: A table and graph that selectively expose the same declared target.

Counterexample: The target itself, treated as identical with its representation.

Projection

A declared transformation from a target description, system state, or semantic assignment into an accessible representational form.

A projection contract specifies:

- target and authority;
- mapping or construction rule;
- interpretation assumptions;
- preserved properties;
- transformed properties;
- omitted properties;
- introduced properties;
- available operations;
- observer assumptions;
- intended use;
- context and scope;
- uncertainty, tolerance, and failure conditions.

Boundary: Projection coordinates established concepts such as views, abstractions, encodings, lenses, simulations, model morphisms, and renderings. It is not a new metaphysical substance.

Canonical ID	projection
Maturity	Proposed analytical role
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	view, encoding, rendering
Deprecated	None
Dependencies	System state, Semantic assignment
Originating paper	Opportunity, Affordance, and Action Availability
Supporting papers	Opportunity, Affordance, and Action Availability; Representation, Invariance, and Projection; Formal Models of Computational Interaction; Computational Interaction Science; Predicting Interaction
Contracts	Projection Contract

Example: A declared mapping from a system state to a GUI, API, or auditory form.

Counterexample: An implementation artifact with no declared target, mapping, or scope.

Scoped invariance

A declared property remains invariant across projections when independently specified comparison rules show that it is preserved for a stated target, observer class, context, metric, and tolerance.

Invariance must be typed. Relevant types include:

- ontological;
- semantic;
- informational;
- structural;
- topological;
- computational;
- inferential;
- task;
- behavioral;
- interactional.

Boundary: No one invariance type implies another.

Canonical ID	scoped-invariance
Maturity	Proposed theory
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	semantic invariance
Dependencies	Projection, Equivalence
Originating paper	Canonical glossary
Supporting papers	None
Contracts	None

Interpretation

The application, recovery, construction, negotiation, or revision of semantic relations by an interpreting system under a context, competence, history, and purpose.

Boundary: Interpretation is a process or event. Semantics specifies or stabilizes admissible relations; interpretation applies or revises them.

Canonical ID	interpretation
Maturity	Proposed
Observable	No - latent/inferred
Latent	Yes
Measurement	Requires latent-variable model
Aliases	None
Deprecated	None
Dependencies	Representation, Interaction history
Originating paper	Ontology, Semantics, Representation, and Interpretation
Supporting papers	Ontology, Semantics, Representation, and Interpretation; Representation, Invariance, and Projection; INTERACTION AS A SCIENTIFIC EXPLANANDUM; Measurement; Computational Interaction Science
Contracts	None

Example: An agent applies conventions and context to a signaled operation.

Counterexample: Formal semantics treated as identical to a participant’s understanding.

Interaction participant

Any bounded person, collective, organism, computational process, artificial system, institution, or hybrid arrangement occupying a consequential role within an interaction.

A participant may observe, interpret, act, constrain, authorize, mediate, represent, store, or transform state. Participation does not imply autonomy, intelligence, consciousness, or learning.

Boundary: Interaction participant is the top-level role. More specific terms such as human participant, computational participant, institution, observer, mediator, autonomous agent, and AI agent should be used when their additional commitments matter.

Canonical ID	interaction-participant
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	participant
Deprecated	None
Dependencies	Agent
Originating paper	Canonical glossary
Supporting papers	None
Contracts	Interaction Contract

Agent

An acting interaction participant that selects or produces actions through a policy, rule, control process, intention, or other action-selection organization.

Boundary: Agent is not the universal term for every participant. An AI model is not automatically an agent; it becomes part of an agentic system when observations, action-selection, and consequential operations are supplied.

Canonical ID	agent
Maturity	Proposed subtype
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	None
Dependencies	None
Originating paper	Opportunity, Affordance, and Action Availability
Supporting papers	Opportunity, Affordance, and Action Availability; Interaction; CAN INTERACTION BECOME A PRIMARY SCIENTIFIC OBJECT?; Formal Models of Computational Interaction; INTERACTION AS A SCIENTIFIC EXPLANANDUM
Contracts	None

Environment

The declared physical, computational, social, symbolic, or mixed system within which action and state transition occur.

Boundary: The environment is not identical with what an agent can observe through a representation.

Canonical ID	environment
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	None
Dependencies	None
Originating paper	Behavior
Supporting papers	Behavior; CAN INTERACTION BECOME A PRIMARY SCIENTIFIC OBJECT?; Interaction; INTERACTION AS A SCIENTIFIC EXPLANANDUM; Ontology, Semantics, Representation, and Interpretation
Contracts	None

Affordance

An action-relevant relation between an agent's effectivities or capabilities and environmental structure that makes a class of action possible. In the Gibsonian sense, perception is not required for the affordance to exist.

Boundary: Affordance is not synonymous with permission, signaling, belief, execution success, or reversibility.

Canonical ID	affordance
Maturity	Established external construct
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	perceived affordance
Dependencies	None
Originating paper	Opportunity, Affordance, and Action Availability
Supporting papers	Opportunity, Affordance, and Action Availability; INTERACTION AS A SCIENTIFIC EXPLANANDUM; Representation, Invariance, and Projection; Ontology, Semantics, Representation, and Interpretation; Predicting Interaction
Contracts	None

Opportunity

A typed and indexed analytical profile coordinating action-availability relations under declared system, agent, projection, task, authority, state, and temporal conditions.

Opportunity is not a new primitive. It coordinates established relations such as feasibility, capability, permission, enablement, reachability, exposure, signaling, belief, expected consequence, reversibility, and uncertainty.

Canonical ID	opportunity
Maturity	Proposed coordination framework
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	opportunity as a primitive
Dependencies	None
Originating paper	Opportunity, Affordance, and Action Availability
Supporting papers	Opportunity, Affordance, and Action Availability; Interaction Similarity; Interaction Strategies; Computational Interaction Science; Predicting Interaction
Contracts	Opportunity Contract

Opportunity profile

For an agent, action, state, projection, and time, an opportunity profile records the relevant typed relations:

- domain-valid;
- effectively capable;
- permitted;
- enabled;
- reachable outcomes;
- exposed;
- signaled;
- believed available;
- expected consequences;
- selected;
- executed;
- successful;
- reversible or recoverable;

- uncertainty and provenance.

Selected, executed, and successful are trace predicates rather than availability sets.

Canonical ID	opportunity-profile
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Operationalizable by contract
Aliases	typed opportunity profile
Deprecated	opportunity as a primitive
Dependencies	Interpretation, System state, Projection, Affordance, Capability, Permission, Enablement, Reachability, Exposure, Signaling, Believed action
Originating paper	Representation, Invariance, and Projection
Supporting papers	Representation, Invariance, and Projection; Opportunity, Affordance, and Action Availability; Behavior; Interaction Similarity; Interaction Strategies
Contracts	Opportunity Contract

Example: A profile distinguishing permitted, enabled, exposed, signaled, and believed actions.

Counterexample: A visible control treated as proof that an action is feasible or believed.

Capability

The effective capacity or resources required for an agent-system coupling to execute an action or bring about an outcome under declared conditions.

Boundary: Capability is distinct from permission. Security capability tokens must be named explicitly when that narrower meaning is intended.

Canonical ID	capability
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	None
Dependencies	None
Originating paper	Opportunity, Affordance, and Action Availability
Supporting papers	Opportunity, Affordance, and Action Availability; Computational Interaction Science; INTERACTION AS A SCIENTIFIC EXPLANANDUM; CAN INTERACTION BECOME A PRIMARY SCIENTIFIC OBJECT?; Interaction
Contracts	None

Permission

Authorization under a policy, role, rule, authority, or institutional condition for a subject to perform an operation.

Boundary: A prohibited action may remain technically possible.

Canonical ID	permission
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	None
Dependencies	None
Originating paper	Opportunity, Affordance, and Action Availability
Supporting papers	Opportunity, Affordance, and Action Availability; Computational Interaction Science; Behavior; Interaction

	Similarity; Interaction Strategies
Contracts	None

Enablement

The condition in which an action's current-state preconditions are satisfied.

Boundary: Enablement is local to the current state and differs from multi-step reachability.

Canonical ID	enablement
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	enabled action
Deprecated	None
Dependencies	None
Originating paper	Opportunity, Affordance, and Action Availability
Supporting papers	Opportunity, Affordance, and Action Availability; Formal Models of Computational Interaction
Contracts	None

Reachability

The existence of at least one admissible transition sequence from a declared initial state to a target state or outcome within specified constraints and horizon.

Canonical ID	reachability
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Operationalizable by contract
Aliases	None
Deprecated	None
Dependencies	None
Originating paper	Opportunity, Affordance, and Action Availability
Supporting papers	Opportunity, Affordance, and Action Availability; Formal Models of Computational Interaction; Interaction
Contracts	None

Exposure

Provision by the current projection of an executable access path to an operation.

Boundary: An exposed operation may be hidden, poorly signaled, unknown, or inaccessible to a particular observer.

Canonical ID	exposure
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Operationalizable by contract
Aliases	operation exposure
Deprecated	None
Dependencies	None
Originating paper	Opportunity, Affordance, and Action Availability
Supporting papers	Opportunity, Affordance, and Action Availability; Interaction Transfer; INTERACTION AS A SCIENTIFIC EXPLANANDUM; Predicting Interaction; Formal Models of Computational Interaction
Contracts	None

Signaling

Perceivable information indicating an operation, condition, consequence, state, or recovery path.

Boundary: A signal may be true, incomplete, ambiguous, or false.

Canonical ID	signaling
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Operationalizable by contract
Aliases	signifier
Deprecated	perceived affordance
Dependencies	None
Originating paper	Opportunity, Affordance, and Action Availability
Supporting papers	Opportunity, Affordance, and Action Availability; Computational Interaction Science; Formal Models of Computational Interaction; Predicting Interaction; Comparative Interaction
Contracts	None

Discoverability

The empirically observed probability and process by which an agent identifies an action or relation under declared task, projection, knowledge, modality, context, and time conditions.

Boundary: Discoverability is not a static visual property.

Canonical ID	discoverability
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Operationalizable by contract
Aliases	None
Deprecated	perceived affordance
Dependencies	None
Originating paper	Opportunity, Affordance, and Action Availability
Supporting papers	Opportunity, Affordance, and Action Availability; Measurement; Representation, Invariance, and Projection; Interaction Transfer; Comparative Interaction
Contracts	None

Believed action

An action represented by an agent as available, with an associated confidence and expected consequence.

Boundary: Belief cannot be read directly from non-selection or task completion. It requires elicitation or a validated inferential model.

Canonical ID	believed-action
Maturity	Proposed latent variable
Observable	No - latent/inferred
Latent	Yes
Measurement	Requires latent-variable model
Aliases	believed availability
Deprecated	perceived affordance
Dependencies	Interaction history
Originating paper	Opportunity, Affordance, and Action Availability
Supporting papers	Opportunity, Affordance, and Action Availability; INTERACTION AS A SCIENTIFIC EXPLANANDUM; Interaction Similarity; Interaction Transfer; Predicting Interaction
Contracts	None

Action

A declared intervention, command, operation, or behavior capable of participating in a state transition.

Boundary: A low-level input such as a click or touch is not necessarily the semantic action under study.

Canonical ID	action
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	perceived affordance, interaction causes behavior
Dependencies	None
Originating paper	Opportunity, Affordance, and Action Availability
Supporting papers	Opportunity, Affordance, and Action Availability; INTERACTION AS A SCIENTIFIC EXPLANANDUM; Interaction Strategies; CAN INTERACTION BECOME A PRIMARY SCIENTIFIC OBJECT?; Behavior
Contracts	Interaction Contract

State transition

A change from one declared system state to another under a transition relation, process, or event.

Canonical ID	state-transition
Maturity	Proposed
Observable	Directly/Instrumentally
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	transition
Deprecated	None
Dependencies	Action
Originating paper	Interaction
Supporting papers	Interaction; Behavior; INTERACTION AS A SCIENTIFIC EXPLANANDUM; CAN INTERACTION BECOME A PRIMARY SCIENTIFIC OBJECT?
Contracts	None

Observation

A recorded or accessible distinction about state, transition, representation, or outcome available through a specified measurement or observation process.

Boundary: Observation is not interpretation, inference, or evidence by itself.

Canonical ID	observation
Maturity	Proposed
Observable	Directly/Instrumentally
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	None
Dependencies	None
Originating paper	Formal Models of Computational Interaction
Supporting papers	Formal Models of Computational Interaction; Measurement; INTERACTION AS A SCIENTIFIC EXPLANANDUM; Interaction Strategies; Interaction Similarity
Contracts	Projection Contract, Opportunity Contract, Interaction Contract, Behavioral Evidence Contract, Measurement Contract, Similarity Contract, Transfer Contract, Strategy Contract, Prediction Contract

Situated behavior

A bounded, temporally extended trajectory of observable or instrumentally recoverable changes, outputs, or regulated persistence attributed to a system in relation to its environment under a declared observation and attribution model.

Boundary: Representation does not determine behavior directly.

Canonical ID	situated-behavior
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	None
Dependencies	None
Originating paper	Canonical glossary
Supporting papers	None
Contracts	None

Behavioral evidence

A provenance-preserving chain connecting direct observations, system-derived features, latent interpretations, alternative explanations, uncertainty, and falsifying evidence to a bounded research claim.

Boundary: A trace is evidence of what occurred. It is not direct evidence of intention, understanding, culture, personality, or protected identity.

Canonical ID	behavioral-evidence
Maturity	Candidate
Observable	Directly/Instrumentally
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	behavior evidence
Deprecated	None
Dependencies	Measurement contract
Originating paper	Behavior
Supporting papers	Behavior; Computational Interaction Science; INTERACTION AS A SCIENTIFIC EXPLANANDUM; Formal Models of Computational Interaction; Comparative Interaction
Contracts	Behavioral Evidence Contract

Behavioral equivalence

Equivalence of declared action or outcome distributions for a specified population, task, context, metric, and tolerance.

Boundary: Behavioral equivalence does not imply semantic, interpretive, strategy, learning, transfer, or interactional equivalence.

Canonical ID	behavioral-equivalence
Maturity	Candidate
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	None
Dependencies	None
Originating paper	Representation, Invariance, and Projection
Supporting papers	Representation, Invariance, and Projection; Interaction Similarity; Formal Models of Computational Interaction; Ontology, Semantics, Representation, and Interpretation; Opportunity, Affordance, and Action Availability
Contracts	Behavioral Evidence Contract

Interactional equivalence

Equivalence of the available operation-transition-observation structure under specified agent, task, projection, authority, state, context, and tolerance conditions.

Canonical ID	interactional-equivalence
Maturity	Candidate
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	None
Dependencies	None
Originating paper	INTERACTION AS A SCIENTIFIC EXPLANANDUM
Supporting papers	INTERACTION AS A SCIENTIFIC EXPLANANDUM; Representation, Invariance, and Projection; Interaction Similarity
Contracts	None

Interaction

A family of bounded coupling phenomena in which distinguishable systems or participants become mutually consequential through perturbation, constraint, uptake, response, and recurrence.

Boundary: Interaction is not reducible to action, behavior, communication, computation, observation, state transition, feedback, or causation alone. Ordinary-language usage may be broader than the strict research definition.

Canonical ID	interaction
Maturity	Proposed scientific target
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	interaction causes behavior
Dependencies	Computational Interaction Science
Originating paper	Interaction
Supporting papers	Interaction; CAN INTERACTION BECOME A PRIMARY SCIENTIFIC OBJECT?; INTERACTION AS A SCIENTIFIC EXPLANANDUM; Computational Interaction Science; Predicting Interaction
Contracts	Interaction Contract

Interaction episode

A bounded interval in which two or more distinguishable systems are coupled such that perturbations originating in, or mediated by, one system alter the possible or actual state trajectory of another, and the resulting configuration constrains a later trajectory of an originating or co-participating system within a declared boundary and timescale.

Reciprocal closure may be direct, indirect, asymmetric, delayed, probabilistic, or environmentally mediated.

Boundary: A single isolated one-way causal effect is not an interaction episode under the strict definition unless it enters a recurrent or reciprocally constraining structure.

Canonical ID	interaction-episode
Maturity	Proposed canonical unit
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Operationalizable by contract
Aliases	strict interaction episode
Deprecated	None
Dependencies	Opportunity profile, Coupling, Token identity, Type identity, Interaction contract
Originating paper	Interaction

Supporting papers	Interaction; CAN INTERACTION BECOME A PRIMARY SCIENTIFIC OBJECT?; Formal Models of Computational Interaction; Computational Interaction Science; Predicting Interaction
Contracts	Interaction Contract

Example: A request, response, and later state-dependent continuation within a declared protocol boundary.

Counterexample: A one-way message with no reciprocal or recurrent constraint inside the declared episode.

Coupling

A relation or pathway through which a participant or shared medium can alter another participant's transition structure, reachable states, probabilities, timing, or constraints.

Canonical ID	coupling
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	None
Dependencies	Interaction participant, Environment
Originating paper	Interaction
Supporting papers	Interaction; CAN INTERACTION BECOME A PRIMARY SCIENTIFIC OBJECT?; INTERACTION AS A SCIENTIFIC EXPLANANDUM; Formal Models of Computational Interaction; How Sciences Define Their Objects
Contracts	None

Perturbation

A difference propagated through a coupling relation that may change state, transition conditions, probabilities, timing, or reachable outcomes.

Canonical ID	perturbation
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	None
Dependencies	None
Originating paper	Interaction
Supporting papers	Interaction; INTERACTION AS A SCIENTIFIC EXPLANANDUM
Contracts	None

Uptake

Incorporation of a perturbation into a participant's state, transition process, control process, or interpreted situation. Uptake need not be semantic or conscious.

Canonical ID	uptake
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	None
Dependencies	None
Originating paper	Interaction

Supporting papers	Interaction; Ontology, Semantics, Representation, and Interpretation; Opportunity, Affordance, and Action Availability
Contracts	None

Recurrence

Return influence through which a response constrains an originating participant, a co-participant, or the shared coupling configuration.

Canonical ID	recurrence
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	None
Dependencies	None
Originating paper	Interaction
Supporting papers	Interaction; Interaction Strategies; Behavior
Contracts	None

Interaction contract

A declaration of interaction identity, participants and roles, boundary, timescale, state, coupling, preconditions, events, transitions, protocol, invariants, evidence, failure, recovery, and completion conditions.

Canonical ID	interaction-contract
Maturity	Proposed reusable artifact
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	None
Dependencies	None
Originating paper	Interaction
Supporting papers	Interaction; Computational Interaction Science; Comparative Interaction
Contracts	Interaction Contract

Interaction morphology

The measurable structural properties of a representation or interaction surface, such as hierarchy, topology, density, alignment, rhythm, symmetry, enclosure, continuity, direction, visual weight, temporal organization, and modality.

Boundary: Morphology is not independent of all semantics or interpretation. A study must declare which content and operations are controlled.

Canonical ID	interaction-morphology
Maturity	Candidate
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	None
Dependencies	None
Originating paper	Canonical glossary
Supporting papers	None
Contracts	Interaction Contract

Interaction grammar

A typed vocabulary and composition structure for semantic actions and transitions independent of any one input modality or rendering implementation.

Canonical ID	interaction-grammar
Maturity	Candidate
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	None
Dependencies	None
Originating paper	CAN INTERACTION BECOME A PRIMARY SCIENTIFIC OBJECT?
Supporting papers	CAN INTERACTION BECOME A PRIMARY SCIENTIFIC OBJECT?; Interaction
Contracts	Interaction Contract

Interaction history

The accumulated experience, training, conventions, outcomes, failures, practices, and prior exposure that may change an agent's later beliefs, policies, and interpretation.

Boundary: Interaction history biases later behavior but does not fully explain or determine an individual.

Canonical ID	interaction-history
Maturity	Candidate
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	None
Dependencies	None
Originating paper	INTERACTION AS A SCIENTIFIC EXPLANANDUM
Supporting papers	INTERACTION AS A SCIENTIFIC EXPLANANDUM
Contracts	Interaction Contract

Trace

A temporally ordered or partially ordered record of observable or attributed events under declared boundaries, granularity, and observation rules.

Boundary: A trace is not identical with behavior, action, policy, strategy, intention, or mechanism.

Canonical ID	trace
Maturity	Proposed
Observable	Directly/Instrumentally
Latent	No or not inherently
Measurement	Operationalizable by contract
Aliases	interaction trace, behavioral trace
Deprecated	None
Dependencies	Observation
Originating paper	Interaction Similarity
Supporting papers	Interaction Similarity; Behavior; Interaction Strategies; Interaction; Predicting Interaction
Contracts	Projection Contract, Opportunity Contract, Interaction Contract, Behavioral Evidence Contract, Measurement Contract, Similarity Contract, Transfer Contract, Strategy Contract, Prediction Contract

Measurement contract

A declaration of the measurand, observation process, operationalization, scale, uncertainty, validity evidence, population, task, projection, comparison conditions, and permitted interpretation.

Boundary: A metric is not a construct. Numerical output does not justify a latent interpretation without a validated measurement relation.

Canonical ID	measurement-contract
Maturity	Proposed reusable artifact
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Operationalizable by contract
Aliases	None
Deprecated	None
Dependencies	Trace
Originating paper	Measurement
Supporting papers	Measurement; Formal Models of Computational Interaction; Computational Interaction Science; Behavior; Interaction Transfer
Contracts	Measurement Contract

Example: A declaration of measurand, observation process, scale, uncertainty, and admissible inference.

Counterexample: A response-time column treated as a construct without validity conditions.

Interaction similarity

A comparison of interaction tokens, types, structures, protocols, opportunities, traces, behaviors, policies, strategies, outcomes, causes, or measurements under an explicit frame, correspondence mapping, admissible transformations, metrics, tolerances, uncertainty, and validity domain.

Boundary: There is no context-free universal similarity score.

Canonical ID	interaction-similarity
Maturity	Proposed typed comparison framework
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Operationalizable by contract
Aliases	None
Deprecated	None
Dependencies	Measurement contract, Similarity contract
Originating paper	INTERACTION AS A SCIENTIFIC EXPLANANDUM
Supporting papers	INTERACTION AS A SCIENTIFIC EXPLANANDUM; Interaction Similarity; Predicting Interaction; Comparative Interaction; Computational Interaction Science
Contracts	Interaction Contract

Example: A typed profile comparing protocol, opportunity, trace, and outcome dimensions.

Counterexample: One aggregate score presented without a comparison contract.

Token identity

Historical identity of one bounded interaction episode under declared continuity, participant, boundary, and timescale criteria.

Canonical ID	token-identity
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	None
Dependencies	None
Originating paper	Interaction

Supporting papers	Interaction; Interaction Similarity; Comparative Interaction
Contracts	None

Type identity

Membership in the same abstraction-defined interaction class under explicit classification rules.

Canonical ID	type-identity
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	None
Dependencies	None
Originating paper	Interaction Similarity
Supporting papers	Interaction Similarity; Interaction; Comparative Interaction
Contracts	None

Equivalence

Exact preservation under a declared comparison contract and observable set. Equivalence is always typed.

Canonical ID	equivalence
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	typed equivalence
Deprecated	None
Dependencies	Interaction similarity
Originating paper	Interaction Similarity
Supporting papers	Interaction Similarity; Representation, Invariance, and Projection; INTERACTION AS A SCIENTIFIC EXPLANANDUM; Comparative Interaction; Formal Models of Computational Interaction
Contracts	None

Similarity

Graded proximity under declared dimensions, representations, alignment rules, metrics, weights, tolerances, and uncertainty.

Canonical ID	similarity
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	typed similarity
Deprecated	universal similarity score
Dependencies	None
Originating paper	Interaction Similarity
Supporting papers	Interaction Similarity; INTERACTION AS A SCIENTIFIC EXPLANANDUM; Computational Interaction Science; Interaction Transfer; Predicting Interaction
Contracts	Similarity Contract

Incomparability

Failure to establish a defensible common comparison space because boundaries, targets, mappings, scales, observation processes, uncertainty, or validity domains cannot be aligned without unsupported assumptions.

Canonical ID	incomparability
Maturity	Proposed
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	None
Dependencies	None
Originating paper	Comparative Interaction
Supporting papers	Comparative Interaction; Interaction Similarity; Computational Interaction Science
Contracts	None

Similarity contract

A declaration of comparison target, frame, representations, correspondence mapping, invariants, variants, observables, admissible and prohibited transformations, typed metrics or predicates, tolerances, uncertainty, validity domain, and final typed profile.

Canonical ID	similarity-contract
Maturity	Proposed reusable artifact
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	None
Dependencies	None
Originating paper	Interaction Similarity
Supporting papers	Interaction Similarity; Predicting Interaction
Contracts	Similarity Contract

Strategy

An inferred, temporally extended organization of conditional action selection, information use, resource allocation, and contingency response that tends to preserve or improve one or more criteria across a declared class of interaction states.

Boundary: A strategy is not a single action, raw trace, repeated sequence, explicit plan, policy alone, skill, habit, or intention. It must be supported through rival discrimination, diagnostic contingencies, uncertainty, and evidence appropriate to the claim.

Canonical ID	strategy
Maturity	Proposed latent explanatory construct
Observable	No - latent/inferred
Latent	Yes
Measurement	Requires latent-variable model
Aliases	None
Deprecated	None
Dependencies	Policy, Interaction history, Strategy contract
Originating paper	Interaction Strategies
Supporting papers	Interaction Strategies; Predicting Interaction; Interaction Transfer; Formal Models of Computational Interaction; Computational Interaction Science
Contracts	Strategy Contract

Example: An inferred pattern of conditional search, commitment, and recovery across episodes.

Counterexample: A single successful action labeled as a strategy.

Strategy contract

A declaration of strategy-bearing system, domain, timescale, criteria, state and observation model, state abstraction, opportunity assumptions, action repertoire, selection policy, information policy, commitments, contingency and recovery rules, switching and stopping conditions, resources, authority, identity invariants, rival strategies, uncertainty, and validity domain.

Canonical ID	strategy-contract
Maturity	Proposed reusable artifact
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	None
Dependencies	None
Originating paper	Interaction Strategies
Supporting papers	Interaction Strategies
Contracts	Strategy Contract

Policy

A conditional relation from represented states, observations, or beliefs to action distributions.

Boundary: A policy may be one component of a strategy. Strategy may additionally include information acquisition, commitment, resource allocation, switching, recovery, and stopping rules.

Canonical ID	policy
Maturity	Proposed
Observable	No - latent/inferred
Latent	Yes
Measurement	Requires latent-variable model
Aliases	None
Deprecated	None
Dependencies	None
Originating paper	Interaction Strategies
Supporting papers	Interaction Strategies; Opportunity, Affordance, and Action Availability; Formal Models of Computational Interaction; Behavior; Interaction Similarity
Contracts	None

Fundamental

A representation-independent computational substrate for declaring domain commitments, evolving system state, assigning operational semantics, and producing governed projections whose preservation, transformation, omission, introduced structure, operations, snapshots, and replay can be inspected.

Boundary: Fundamental is not the scientific theory, a philosophical ontology, proof of CIS, or a guarantee of shared human meaning.

Canonical ID	fundamental
Maturity	Proposed research substrate
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	None
Deprecated	None
Dependencies	None
Originating paper	CAN INTERACTION BECOME A PRIMARY SCIENTIFIC OBJECT?
Supporting papers	CAN INTERACTION BECOME A PRIMARY SCIENTIFIC OBJECT?; How Sciences Define Their Objects; Representation, Invariance, and Projection; Predicting Interaction
Contracts	None

Computational Interaction Science

A research program that studies interaction as an explanatory target by separating and comparing declared domain commitments, state, semantics, projection, representation, interpretation, opportunity, coupling, behavior, measurement, similarity, strategy, and evidence.

CIS must justify itself through comparative explanatory gain, empirical validity, reproducible methods, and error-sensitive predictions. It does not replace HCI, psychology, anthropology, cognitive science, design, computer science, or related disciplines.

Canonical ID	computational-interaction-science
Maturity	Proposed research program; disciplinary autonomy unproven
Observable	Conditional on observation model
Latent	No or not inherently
Measurement	Measure only under declared operationalization
Aliases	CompInt
Deprecated	None
Dependencies	None
Originating paper	Computational Interaction Science
Supporting papers	Computational Interaction Science; Predicting Interaction
Contracts	None